

Different cells, different models: choosing a modelling paradigm by cell biology

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The current discourse on the AI virtual cell, exemplified by the Chan Zuckerberg Initiative's Virtual Cells Platform (1), is dominated by transcriptomics foundation models trained and validated on dividing, nucleated, well-sampled cell lines such as K562, HEPG2, RPE1, and Jurkat. Mechanistic whole-cell modelling (2, 3) is frequently positioned as a historical predecessor rather than a complementary paradigm. We argue that the choice of modelling paradigm should be driven by the cell biology and the research question, not by technology momentum. We compare three paradigms — mechanistic whole-cell models, transcriptomics foundation models, and hybrid causal-mechanistic machine learning — on what each does well, what each does poorly, and how validation and computational cost differ. We use the human platelet — anucleate, terminally differentiated, with a circulating lifespan of approximately nine days and signalling events on sub-minute timescales — as a worked example that breaks the transcriptomics-foundation-model orthodoxy and makes the case for mechanistic and hybrid approaches. We close with three pragmatic convergences: machine-learning priors for mechanistic parameters, mechanistic models as training signal for learned representations, and AI surrogates that accelerate mechanistic simulation.

whole-cell modelling | virtual cell | foundation models | platelet | mechanistic biology

Introduction

The phrase *AI virtual cell* (AIVC) has come, in the space of two years, to dominate the public face of computational cell biology (1, 4, 5). The Chan Zuckerberg Initiative's Virtual Cells Platform, the recent crop of single-cell foundation models (6–8), and a growing body of position pieces from both academia and industry (9, 10) frame the virtual cell as a learnable object: a representation, trained on many millions of single-cell transcriptomes, capable in principle of generalising across tissue, perturbation, and species. The framing is seductive, and the engineering achievements are real. It is also a framing that, almost everywhere it is articulated, assumes a particular set of cells. The cells in question divide; they are nucleated; their transcriptional state is informative about their functional state on the timescale of measurement; and they are abundant enough in public single-cell atlases that a foundation model has something to fit. K562, HEPG2, RPE1, Jurkat, and the rest of the proliferating-cancer-line canon are the implicit referents whenever a virtual-cell paper reports a benchmark.

This is unobjectionable so long as those cells exhaust the biology one cares about. They do not. A great deal of cell biology is done outside the dividing-and-nucleated canon. Terminally differentiated neurons, cardiomyocytes, red blood cells, and platelets together account for a substantial fraction of the cells in the human body and a similarly large fraction of clinically relevant biology, yet none of them fit the implicit transcriptomics-first frame: neurons and cardiomyocytes are post-mitotic and exhibit functional dynamics — action potentials,

Significance Statement

The AI virtual-cell programmes that now dominate computational cell biology are built on a single methodological wager: that learned representations of single-cell transcriptomes will generalise to arbitrary cells. That wager pays out well for the cells the field has trained on — proliferating, nucleated, transcriptionally active human cancer lines. It pays out badly for cells whose biology cuts against the dominant validation set: anucleate cells, terminally differentiated cells, cells whose interesting dynamics happen on sub-minute timescales, and cells for which single-cell data are sparse. By foregrounding the cell biology before the modelling technology, we argue that mechanistic and hybrid approaches are not historical predecessors of AI virtual cells but co-equal paradigms whose use should be determined by the cell and the question.

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excitation-contraction coupling — on timescales where transcription is silent. Red blood cells have no nucleus. Platelets have no nucleus and no organised transcriptional apparatus to speak of, complete their biological work over seconds to minutes, and are catastrophically under-represented in single-cell atlases relative to their numerical abundance in blood. For these cells, the methodology one chooses is not a matter of taste; it is forced by the cell.

The argument of this Perspective is that the choice of modelling paradigm should follow from the cell biology and the research question, not from the prevailing technology. We make that argument by contrasting three modelling paradigms now in active use, locating each on a small set of axes that capture the practical trade-offs between them, and then working through the human platelet as a case in which the dominant transcriptomics-foundation-model paradigm breaks down and mechanistic and hybrid approaches succeed.

Three modelling paradigms

Mechanistic whole-cell models. Mechanistic whole-cell modelling, exemplified by the *Mycoplasma genitalium* model of (author?) (2) and the *Escherichia coli* WCM of (author?) (3), takes the position that a cell is a system of identifiable biochemical processes — transcription, translation, replication, metabolism, signalling — each with a known mechanism and known rate constants, coupled through a shared state of molecular counts and concentrations. A typical whole-cell model carries explicit kinetic equations for thousands of reactions [TODO confirm gene/reaction counts for Karr 2012 (~525 genes / 28 processes) and Macklin 2020 (~4 500 genes / 31 processes)], with simulation proceeding by alternating process-resource-request, state-partition, and process-evolve steps under a global timestep. The contracts between processes and shared state are typed and explicit: any parameter that enters a rate expression is traceable to a data source through what the wcEcoli literature calls the *sim_data* mapping (3). Validation proceeds in two directions, against system-level phenotypes (growth rate, doubling time, gene-knockout viability) and against subsystem quantitative behaviour (metabolic flux maps, ribosome occupancy).

What a mechanistic whole-cell model does well is that it forces every quantity in the model to be either measured, derived, or explicitly calibrated, and it makes the reasons for a particular simulated behaviour interpretable in the language of the underlying biology. What it does badly is scale and data hunger: building a mechanistic whole-cell model requires substantial domain expertise per process, careful curation of rate constants, and a runtime cost that does not trivially shrink with hardware. The result is a small number of flagship models, each representing years of work, and an order-of-magnitude gap between

the mechanistic detail accessible to modellers and the catalogue of cells biologists would like to study.

Single-cell transcriptomics foundation models. The second paradigm replaces mechanism with learned representation. A transformer or related deep architecture is trained on tens to hundreds of millions of single-cell transcriptomes — Geneformer (6) on approximately 30 million cells [TODO confirm], scGPT (7) on approximately 33 million [TODO confirm], and the most recent CZI TranscriptFormer release on approximately 112 million cells across 12 species [TODO confirm exact figure] — and acquires a latent space in which cell type, state, and perturbation response can be read out by lightweight downstream heads. Foundation models excel at three tasks: zero- or few-shot cell-type annotation across heterogeneous tissues, in-silico perturbation prediction within the regime they have been trained on, and the generation of representations that improve the data efficiency of downstream tasks such as drug-target identification. The CZI Virtual Cells Platform (1) and the recent rBio system (11) formalise these capabilities as the operating infrastructure of an emerging virtual-cell programme.

What this paradigm does well is the breadth that mechanism cannot afford: a foundation model trained on tens of millions of cells will have *seen* far more biological context than any mechanistic model will ever encode. What it does poorly is the kind of behaviour that is not in the training distribution: cells with limited single-cell representation, dynamics that unfold on timescales shorter than the transcriptional snapshot, and any setting where the question is *why* something happens rather than *what* happens. Validation is benchmark-driven and intrinsically retrospective: a foundation model that has not been measured against a held-out perturbation cannot be trusted on it. The computational footprint is concentrated in training rather than inference; inference is cheap, training is at the limit of what academic groups can afford.

Hybrid and causal-mechanistic ML. A third paradigm — hybrid in the strict sense — augments mechanistic models with learned components, or constrains learned models with mechanistic structure. The Valence Labs position paper (9) frames this as *predict, explain, discover*: prediction from learning, explanation from mechanism, discovery from the interaction. Concrete instances include physics-informed neural networks for ODE systems, neural ODE surrogates that approximate the output of expensive mechanistic simulations at a fraction of the runtime cost, and mechanism-anchored language models that use a kinetic-model black box as either a training signal or a verifier (the rBio approach (11)). The “LLMs Meet Virtual Cell” survey (12) catalogues a growing list of such hybrids.

What hybrid approaches do well is to address the failures of either pure paradigm: AI provides scale and pattern-recognition that mechanism cannot match; mechanism provides causal interpretability and out-of-distribution validity that pure learning cannot. What they do poorly is methodology — the field is young, the engineering norms are unsettled, and the validation regimes for hybrids are an open problem. Computational footprints span the full range, from neural ODE surrogates that are cheaper at inference than the simulation they replace, through to physics-informed networks whose training cost matches that of the foundation models themselves.

A taxonomy by cell biology and question

A paradigm choice can be located along three axes that together capture the practical trade-off space: *mechanism explicitness*, the degree to which the model's behaviour can be traced to identifiable biological reactions; *data dependence*, the volume and density of empirical observations required to fit the model; and *interpretability requirement*, the extent to which the user requires the model's predictions to come with a causal account. Figure 1 sketches this space and locates each paradigm in it.

These axes are not orthogonal in practice, but they map cleanly onto the questions a working biologist actually asks. *What state is this cell in?* is a classification question well-served by foundation models. *How will this perturbation propagate?* spans both paradigms depending on whether the perturbation is in or out of the training distribution. *Why does this cell make this decision?* is a causal question that only mechanism can answer cleanly, however much correlation a learned model uncovers. And — the question we develop in the next section — *what does this cell do on a sub-minute timescale when its transcriptional state is mute?* is a question for which mechanism is not just preferable but necessary.

The platelet as a worked example

The human platelet is the cell that breaks the dominant paradigm. It is anucleate — derived as a cytoplasmic fragment from a megakaryocyte, with no genome of its own and a residual, non-canonical transcriptome inherited at the moment of release [TODO confirm best citation for platelet transcriptome residuality; candidates include Rowley et al. 2011, Schubert et al. 2014]. It is terminally differentiated, with a circulating half-life of approximately nine days and no division. Its defining functional behaviour — adhesion, activation, granule secretion, aggregation, and procoagulant transformation — unfolds on timescales of seconds to minutes after vascular injury, governed by calcium signalling, integrin reorganisation, and granule release rather than by changes in mRNA abundance. And it is poorly represented in public single-cell transcriptomic atlases relative to its abundance in the bloodstream, in part because of

Figure 1 sketch. Paradigm placement along three axes:

mechanism explicitness (low → high), *data dependence* (low → high), *interpretability requirement* (low → high).

Whole-cell mechanistic models live at the high-mechanism, low-data, high-interpretability corner.

Transcriptomics foundation models live at the low-mechanism, high-data, low-interpretability corner.

Hybrid approaches occupy the interior diagonal.

Figure 1. Paradigm classification by cell biology and question. The three modelling paradigms are placed in a three-axis space defined by mechanism explicitness, data dependence, and interpretability requirement. The axes capture the practical trade-offs that drive paradigm selection in working cell biology. (Final figure: TikZ / mermaid sketch to replace the placeholder box.)

methodological difficulties in recovering anucleate cells in standard droplet-based workflows.

Each of these properties closes off a different affordance of the transcriptomics-foundation-model paradigm. *Anucleate*: there is no ongoing transcriptional response to model, only a fixed residual mRNA pool that does not change in any informative way on the timescale of activation. *Terminally differentiated*: the perturbation-response benchmarks that foundation models are validated against, where the response is mediated by transcriptional reprogramming, are not the biology of interest. *Sub-minute dynamics*: the model time-resolution required to capture calcium transients, integrin inside-out signalling, and granule release is incompatible with single-cell transcriptomic measurement. *Sparse data*: even where one would like a learned representation, the volume of single-platelet data required to fit a foundation model is not available, and may not be obtainable with current technology.

By contrast, mechanistic modelling is well-suited to all four properties. The platelet whole-cell model under construction in our group [TODO add citation / link to repository / preprint if available] extends the wcEcoli engine to a single platelet, with explicit processes for calcium dynamics, granule secretion, integrin activation, and metabolism, and explicit compartments for the cytosol, the dense tubular system (the platelet's calcium store), and the dense and alpha granules. Validation targets are biophysical: the cytosolic calcium transient measured by (author?) (13) under thrombin / SFLLRN stimulation; the steady-state ATP turnover measured by (author?) (14, TODO confirm citation, 1979 or 1981); the SOCE flux under EDTA / no-Ca² conditions; the calcium-pump and IP3-receptor activity reported in the

Figure 3 sketch. Platelet case study, paradigm fit by question.

Rows: (i) transcriptomic state, (ii) perturbation response, (iii) Ca^{2+} dynamics, (iv) granule secretion, (v) integrin activation.

Columns: foundation-model fit, hybrid fit, mechanistic fit.

Shading: green = paradigm well-suited; amber = partial; red = paradigm cannot address.

Expected pattern: rows (iii)–(v) red under foundation model, amber under hybrid, green under mechanistic.

Figure 2. Platelet case study by paradigm. Matrix of platelet biological questions against modelling paradigm, with paradigm fit shaded by suitability. The mechanistic column dominates on the biophysical-dynamics rows; the foundation-model column collapses because the cell's defining biology is not transcriptional. (Final figure: full matrix with literature anchors per cell to be drafted.)

detailed kinetic literature (15). None of these targets is the sort of data foundation models are built to consume.

A hybrid approach has a role to play in this setting, but only after the mechanistic scaffold is in place. Learned components could supply priors for poorly-constrained rate constants — and the proteomic correlation observed by (author?) (16) between SERCA and IP3R copy numbers across donors is exactly the kind of empirical relationship that a small learned model might encode as a structured prior. ML surrogates could accelerate the calcium-signalling ODE for ensemble simulation, particularly important once stochastic effects in the low-copy receptor cascade are added (a regime in which one wishes to simulate hundreds of platelet replicates rather than one). None of this displaces the mechanistic core; it amplifies it.

A software-engineering lens

A complementary view of the same trade-offs comes from software engineering rather than biology, and it is a view the AIVC literature has under-developed. A mechanistic whole-cell model is, structurally, a large, typed, contract-bound piece of software. Every reaction has a named rate constant, every rate constant has a source citation, every process advertises its molecular requests in a partition contract, and every state variable is identifiable in the simulation output by compartment-tagged species name. This is the *sim_data* discipline of the wcEcoli lineage (3), and it imposes a non-trivial cost: each new process is a code review, a citation audit, and an integration test against the rest of the model. The payoff is that any simulated behaviour is interpretable in the language of the underlying biology, and any disagreement with experiment is locatable in a specific rate constant or process.

A foundation model is structurally different. The model is a set of weights; the *sim_data* equivalent is the training corpus; the contract is statistical rather than typed. Reproducibility costs lie not in code review but in dataset versioning and seed management. Validation costs are concentrated in benchmark curation rather than mechanism audit. Modularity, where it exists, is the modularity of LoRA adapters and downstream heads rather than the modularity of cellular processes. None of this is wrong — it is the appropriate engineering posture for a model whose value derives from the breadth of training data — but it means that the kinds of guarantees one can extract from a foundation model are statistically scoped in a way that a mechanistic model's guarantees are not.

This distinction matters when the use case is regulatory, clinical, or adversarial. A pharmaceutical perturbation prediction that comes with an interpretable causal account of *why* the drug affects the cell this way is a defensible artefact in a way that a predicted transcriptomic shift, however accurate, is not. The Karr 2012 model demonstrated this stance with a gene-essentiality screen (2) in which every essentiality call was traceable to a specific process — a property no foundation-model benchmark currently delivers.

Convergence: priors, training signal, surrogates

The framing of paradigm choice as a binary is misleading; the practically interesting question is how the paradigms compose. Three specific convergences are already legible in the recent literature and are the natural near-term scope for any platelet-modelling or adjacent-cell programme.

First, *AI models as priors for mechanistic parameters*. A mechanistic whole-cell model has hundreds to thousands of rate constants, many of them constrained only weakly by experiment. A learned model trained on proteomic, transcriptomic, and pathway databases is in a position to supply structured priors over those rate constants — encoding, for instance, the cross-cell-line correlations that (author?) (16) observe between SERCA and IP3R copy numbers, or the broader pump-receptor stoichiometric patterns that proteomic atlases now make visible. The mechanistic model remains the object of inference; the prior is the contribution.

Second, *mechanistic models as training signal*. The rBio approach (11) uses kinetic models as a verifier for a large language model's perturbation-response predictions: predictions that are inconsistent with mechanism are penalised. The pattern generalises. Any mechanistic model fast enough to evaluate at training time can serve as a constraint on a learned model, narrowing the hypothesis class to mechanistically defensible regions.

Figure 2 sketch. Decision matrix for paradigm selection.

Rows: (i) cell biology has rich single-cell data; (ii) cell biology has sub-minute dynamics; (iii) question is causal-explanatory; (iv) question is perturbation-prediction within training distribution; (v) regulatory / clinical interpretability required.

Columns: foundation model recommended; hybrid recommended; mechanistic recommended.

A clinician answering (i) no, (ii) yes, (iii) yes, (iv) no, (v) yes lands squarely in the mechanistic column.

Figure 3. Decision matrix for paradigm selection. Cell-biology and question-type axes against recommended paradigm. The matrix is deliberately coarse: it expresses where each paradigm is well-suited, not a quantitative model-selection procedure. (Final figure: full matrix with literature exemplars per cell to be drafted.)

Third, *AI surrogates for accelerated mechanistic simulation*. A neural ODE or transformer surrogate trained on the outputs of a mechanistic simulation can deliver inference-time speeds two to four orders of magnitude faster than the underlying integrator, at the cost of out-of-distribution validity. For ensemble simulation — indispensable once stochasticity is added to a low-copy signalling cascade — this is the difference between a single replicate per hour and a thousand. Figure 2 sketches the decision space.

Conclusion

The argument of this Perspective is not that the AI virtual cell is a mistake, nor that mechanistic whole-cell modelling is a recovered orthodoxy. It is that the choice between them — and the choice of any hybrid between them — is properly made on the basis of the cell and the question, not on the basis of technology momentum. The human platelet is one cell among many for which the dominant paradigm's assumptions break, and for which a mechanistic and hybrid stance is forced by biology. Other cells — post-mitotic neurons, cardiomyocytes, erythrocytes, terminal effector lymphocytes — pose their own variants of the same problem. Foundation models, mechanistic models, and hybrids each have a clear remit; that remit is determined by the data, the dynamics, the interpretability requirement, and the question, not by which paradigm has the largest current grant portfolio. We close, deliberately, with no methodological consensus beyond that: fit the paradigm to the cell.

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Author contributions

S.H. conceived the argument and drafted the manuscript. M.F. TODO.

Conflict of interest

The authors declare no conflicts of interest.

1. C Bunne, Y Roohani, et al., How to build the virtual cell with artificial intelligence: priorities and opportunities. *Cell* **0**, 0–0 (2024) TODO confirm exact author list, volume, issue, page range, and DOI; published late 2024.
2. JR Karr, et al., A whole-cell computational model predicts phenotype from genotype. *Cell* **150**, 389–401 (2012) Mycoplasma genitalium; 525 genes / 28 processes — TODO confirm exact gene/process counts.
3. DN Macklin, et al., Simultaneous cross-evaluation of heterogeneous *e. coli* datasets via mechanistic simulation. *Science* **369**, eaav3751 (2020) *wcEcoli*; TODO confirm exact gene count (4 558) and process count (31).
4. A Cui, et al., Multimodal foundation models for the virtual cell: a perspective (Perspective / preprint) (2025) TODO confirm exact venue, author list, title (working title used here), DOI.
5. A Yu, et al., Ai-driven virtual cell models in preclinical research. *npj Digit. Medicine* **0**, 0–0 (2025) TODO confirm exact author list, volume, issue, page range, DOI.
6. CV Theodoris, et al., Transfer learning enables predictions in network biology. *Nature* **618**, 616–624 (2023) GENEFORMER; TODO confirm exact pretraining corpus size (30M cells).
7. H Cui, et al., scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nat. Methods* **21**, 1470–1480 (2024) scGPT; TODO confirm exact training corpus size (33M cells).
8. Chan Zuckerberg Initiative, Transcriptformer: a cross-species single-cell transcriptomics foundation model (Technical report / model card) (2025) TODO confirm exact title, citation form, training corpus size (user reported 112M cells across 12 species).
9. Valence Labs, Virtual cells: Predict, explain, discover (Valence Labs position paper) (2025) Company position paper; not peer-reviewed. TODO confirm exact title, URL, publication date.
10. A Szalata, et al., Transformers in single-cell omics: a review and new perspectives. *Nat. Methods* **0**, 0–0 (2024) TODO confirm exact title, volume, issue, page range, DOI; published mid-2024.
11. Chan Zuckerberg Initiative, rbio: mechanism-anchored reasoning for the virtual cell (Technical report / preprint) (2025) TODO confirm exact title, venue, authorship, URL/DOI.
12. Various, Lims meet virtual cell: a survey of language models in cellular modelling (arXiv preprint) (2025) TODO locate exact arXiv ID, authors, title — this is a working title from the user prompt.
13. AT Dolan, SL Diamond, Systems modeling of Ca^{2+} homeostasis and mobilization in platelets mediated by IP3 and store-operated Ca^{2+} entry. *Biophys. J.* **106**, 2049–2060 (2014).
14. H Holmsen, Platelet metabolism and activation (TODO) (1981) TODO confirm exact reference — candidates: Holmsen 1979 *Thromb. Haemost.* or Holmsen 1981 *Semin. Hematol.*
15. JE Purvis, MS Chatterjee, LF Brass, SL Diamond, A molecular signaling model of platelet phosphoinositide and calcium regulation during homeostasis and P2Y1 activation. *Blood* **112**, 4069–4079 (2008).
16. FA Balabin, JDD Korobkina, SV Galkina, MA Panteleev, AN Sveshnikova, Personalization of a computational systems biology model of blood platelet calcium signaling. *Biomeditsinskaya Khimiya* **70**, 394–402 (2024).